

NOOR MOHAMMADI

Software Engineer · Cloud & AI Systems · Full-Stack · Robotics & IoT · Game Developer · Data Annotator



noor3moha67@gmail.com · [Go to my Github](#) · [Go to my Portfolio](#) · [Go to my Linked-In](#)

PROFESSIONAL SUMMARY

I am a Passionate and Proactive person who is always striving to put my work into the world. 4× first-place hackathon winner (IEEE conclave IBM, ITMO University Russia, Government level). Published preprint researcher with one of the research papers accepted. Active contributor to LLM RLHF evaluation pipelines at scale. Built Firebase-backed live web applications, ML models deployed in Docker and Google Colab, Unity 3D games, Unreal Engine Games and real-time 3D rover simulation using Three.js / Node.js with real-time data stream to Firebase. Currently freelancing at Deccan AI Experts (Mar 2026–present) on AI-driven solutions. Seeking to bring a blend of accuracy and versatility to my work.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), C, C++, C#, Java (familiar), SQL, Embedded C, GDScript
Cloud & APIs	Google Cloud Platform (GCP), Firebase (Realtime DB, Hosting), AWS (S3, Lambda concepts), REST API design, Flask microservices, ThingSpeak, Blynk, webhook workflows
AI / ML	Gemini API, LLM prompt engineering, RLHF evaluation, scikit-learn, ML model deployment (Docker, Colab), OpenCV, Gen AI workflow automation, Data Annotation with Attention to detail.
Data Engineering	Pandas, NumPy, SciPy, Matplotlib, automated telemetry pipelines, real-time sensor aggregation, anomaly detection
Web & Front-End	HTML5, CSS3, Tailwind CSS, jQuery, DOM manipulation, Three.js, Node.js, responsive design, Firebase Hosting
Embedded Systems	ESP32, ESP8266, Raspberry Pi, Arduino IDE, I2C, SPI, UART, PWM, GPIO, real-time control loops
Databases	MySQL (schema design, normalised queries), Firebase Realtime Database
Dev Practices	Git/GitHub, feature branching, code review, unit & integration testing, agile sprints, technical documentation, Docker
Platforms & Tools	VS Code, Linux, VMware, Unity 3D, Unreal Engine 5.5, Godot 4, Google Colab, Microsoft office suite (Excel, Word).

WORK EXPERIENCE

AI Data Annotator — Freelance | [Deccan AI Experts](#)

Mar 2026 – Present

Remote

- ▶ Assisting in human-in-the-loop workflows for AI Training, such as Data Annotation and RLHF Preference Ranking.
- ▶ Evaluate and Assess the quality of AI model response in multiple rubrics from tool execution (JSON) to user experience and task fulfillment across multi-turn and single-turn conversations with context.
- ▶ Used my best Research, Technical and such strong writing skills acquired from many years of Independent freelance work and collaborated academic experiences to demonstrate superior quality and analysis of AI model responses and Annotation.

IoT Intern — Embedded Software & Cloud Integration | [Innovate Intern](#)

Jun–Jul 2025

Remote

- ▶ Designed and implemented ESP8266 firmware (C/C++) for sensor acquisition, Wi-Fi provisioning, and real-time control loops — reduced sensor data latency ~20% across 3 deployed prototypes.
- ▶ Built end-to-end edge-to-cloud integration pipelines via ThingSpeak REST APIs; engineered HTTP request handlers achieving <2s end-to-end update intervals.
- ▶ Validated 100% of firmware modules through systematic unit testing; authored integration documentation covering API specs, pin mappings, and system architecture.
- ▶ Defined software-hardware interfaces with the hardware team, reducing integration issues by 50%.

Front-End Intern — Software Development & Client Integration | [Unified Mentor](#)

Jul–Aug 2025

Remote - Coimbatore, India

- ▶ Delivered 5+ modular features using HTML5, CSS3, Tailwind CSS, and JavaScript within agile sprint cycles, meeting 100% of sprint deadlines.
- ▶ Translated requirements from 3+ client engagements into software specifications; performed cross-browser regression testing across 4 environments.
- ▶ Managed feature branching, commits, and pull requests in a shared Git repo; code reviews reduced bug backlog ~30%.

AI Prompt Engineer & Evaluator — Freelance | [Outlier](#)

Jun 2025

Remote

- ▶ Evaluated and rated 200+ Gen AI-generated outputs for correctness, instruction-following, and safety — directly contributing to LLM RLHF fine-tuning pipelines.
- ▶ Developed systematic analysis frameworks for identifying Gen AI failure modes across 10+ task categories.

Freelance Creative and Technical Writer | [Independent](#)

Mar 2021 – Dec 2024

Remote

- ▶ Produced structured technical documentation, blogs and ghost writing of creative poetry and fictional literature for 3+ years according to client requirements over various platforms to overseas clients.

PROJECTS

Aether — Open-World Horror RPG (Unity 3D)

Apr 2026

Unity 3D, C#, game AI systems, OOP, Git

- ▶ Designed and developed a full open-world 3D horror RPG in Unity using C# — architected core gameplay systems, AI NPC state machines, and physics interactions.

- Applied software engineering practices: modular OOP design, version control via Git, systematic debugging.

Mars Rover — Live Firebase Cloud Platform with 3D Simulation & ML

Mar 2026

ESP32, Embedded C/C++, Flask, Firebase Realtime DB, Three.js, Node.js, HTML/CSS, scikit-learn, OpenCV

- Built a Firebase Realtime Database-backed cloud platform tracking live rover telemetry — deployed a 3D Three.js / Node.js simulation that mirrors real rover position and sensor state in real time.
- Engineered firmware for a 6-wheeled ESP32-based rover: DC motor control algorithms, PWM timing, real-time sensor loops; integrated ESP32-CAM for computer vision obstacle detection.
- Designed Flask REST API cloud control layer enabling full remote command dispatch from Android and desktop clients with <500ms round-trip latency.
- Built automated ML telemetry pipeline processing 10,000+ AQI, image, and multi-sensor data points; applied scikit-learn anomaly detection achieving 92%+ accuracy and OpenCV real-time analysis at 30 FPS.
- All projects hosted on GitHub and Firebase Hosting: github.com/Mai-Xio/CloudAIQud

AgroSense — Vertical Farm AI Platform (Firebase, Docker, ML, Live Sensors)

Mar 2026

Python, scikit-learn, Firebase Realtime DB, Google Colab, Docker, ESP32, HTML/CSS

- Developed a live vertical farm monitoring web app hosted on Firebase (mai-xio.github.io/AgroSense/) pulling real sensor data from ESP32 nodes.
- Built and deployed ML models in Google Colab and locally via Docker to classify crop health and forecast yields from live IoT sensor streams.
- Wrote embedded C sensor firmware for the vertical farm (temperature, humidity, light, CO₂) with cloud push to Firebase Realtime DB.
- Full stack: Firebase Hosting front-end → Firebase Realtime DB → Python ML backend → Docker local inference.

MedFuseNet — Automated Fracture Detection ML Pipeline

Mar 2026

Python, OpenCV, NumPy, Pandas, scikit-learn, SciPy, deep learning

- Engineered automated medical image analysis pipeline for bone fracture detection — processed 1,000+ X-ray images with 40% preprocessing speedup via batched NumPy operations.
- Designed evaluation automation with scikit-learn (precision, recall, F1) and Matplotlib dashboards across 3+ dataset splits.

Loop Breaker — Sci-Fi First-Person Shooter (Unreal Engine 5.5)

Jan 2026

C++, Blueprint Visual Scripting, Unreal Engine 5.5, Gemini API, ML integration

- Architected core FPS gameplay loop in C++ and Blueprints — designed player controller, weapon systems, and game state management from scratch.
- Integrated Gemini API and ML models for adaptive enemy behaviour — built NPC decision trees responding dynamically to player actions.
- Engineered post-processing visual pipeline with UE5 Lumen global illumination and custom environmental storytelling assets.

Spider Cat Quadruped Robot — Gen AI Cloud Integration Backend

Nov 2025

Python, Flask, REST API, Gemini API, ESP8266, Embedded C, JSON, unit testing

- Architected a Flask REST API backend integrating ESP8266 edge hardware with Google Gemini Gen AI — designed full request/response JSON schema, handled streaming AI responses, optimised for <300ms command latency.
- Built a Gen AI automation loop: sensor inputs trigger AI inference, which dispatches control commands back to hardware.
- Implemented unit tests covering 8+ API endpoints and firmware command handlers; eliminated hardware-cloud sync errors before final demo.
- Won 1st Place — IEEE Conclave IBM Project Pitching (LPU, Nov 2025).

Dino Runner — 2D Platformer (Python / Pygame)

Nov 2025

Python, Pygame, 2D asset creation

- Built a complete 2D runner game end-to-end — designed all sprites, audio, and game logic in Python with Pygame.

- ▶ Implemented 3-tier difficulty scaling (Easy / Medium / Hard) with procedurally increasing obstacle speed and spawn rate.

Submerged — First-Person Survival Game (Unreal Engine 5.5) | Team Nyx

Aug 2025

C++, Blueprint Visual Scripting, UE5 Lumen, Nanite, inventory systems

- ▶ Led visual and rendering direction for a full underwater survival experience — implemented Lumen GI, volumetric atmospheric effects, and custom underwater post-processing.
- ▶ Designed and implemented core survival systems: inventory management, resource tracking, and environmental hazard logic.
- ▶ Won 1st Place + Best Visual + Best Gameplay — ITMO University GameJam (international competition, Aug 2025).

Muhaana & Tulipa — Full-Stack Web Platforms with Firebase & MySQL

Aug 2025

HTML5, CSS3, JavaScript, jQuery, MySQL, REST integration, Firebase

- ▶ Deployed 2 production web platforms with MySQL and Firebase database integration — normalised schemas and optimised SQL queries supporting 500+ product listings and user records.
- ▶ Implemented dynamic DOM manipulation, jQuery event handling, and responsive layouts; validated across 4+ browser and device environments.

EDUCATION

B.Sc. Information Technology (2023-26) | CGPA: 8.70

Aug 2023 – June 2026

Lovely Professional University, Phagwara, Punjab

ACHIEVEMENTS & RECOGNITION

- ▶ Paper Acceptance for the title “Hybrid Autonomous Ground-Air Inspection Rover with Integrated Drone Docking, Obstacle Detection, Image Analysis, and AQI-Based Machine Learning for Hazardous Urban Environment Monitoring” at **Indian Journal of Computer Science and Technology (IndJCST)** under ISSN 2583-5300 with Paper ID: INDJCST-02454 May 2026
- ▶ Represented LPU at [Innovate India](#) Government Hackathon | Bharat Mandapam, New Delhi Mar 2026
- ▶ Speaker / Attendee — [South Asian Technology Conference](#) Mar 2026
- ▶ Runner-Up — [National Level GameJam Hackathon](#) | IIT Ropar Feb 2026
- ▶ 1st Place — IEEE Conclave IBM Project Pitching (4 embedded/software projects) | LPU Nov 2025
- ▶ 1st Place + Best Visual + Best Gameplay — [ITMO University International GameJam](#) - [Read the web article on our win](#) | Team Nyx Aug 2025
- ▶ 1st Place — Battle of Wits 9.0 SDG Ideathon | LPU Mar 2025
- ▶ 1st Place — Interview Strategist 5.0 Competitive Bootcamp | LPU Mar 2025
- ▶ Preprint (Aug 2026, forthcoming): Research paper submission pending publication
- ▶ Preprint: 'Systematic Review and Analysis of AI Chatbots in Customer Service' - DOI: [10.13140/RG.2.2.21894.59206](https://doi.org/10.13140/RG.2.2.21894.59206) Nov 2024
- ▶ Preprint: 'Fairness in CPU Scheduling: A Systematic Analysis' - DOI: [10.13140/RG.2.2.34477.50402](https://doi.org/10.13140/RG.2.2.34477.50402) Nov 2024

CERTIFICATIONS

- ▶ Remote Sensing Data Analytics for Crop Production Forecasting — ISRO / IIRS, Govt. of India | Grade: A+ Jun–Jul 2025

C++ Programming for Unreal Game Development (4-course Specialisation) — University of Colorado, Coursera

Jul 2025

‣ DU Lab Summer School, GameDev Track — ITMO University

Aug 2025

‣ Advanced Diploma in Software Engineering — IISDT

Aug 2023

‣ Web Development Crash Essentials — Mind Luster

Nov 2024

‣ Operating Systems — Saylor Academy

Apr 2024